

KTi₂(PO₄)₃ Electrode with a Long Cycling Stability for Potassium-Ion Batteries

Natalia Voronina, Jae Hyeon Jo, Aishuak Konarov, Jongsoon Kim,*
and Seung-Taek Myung*

In this work, rhombohedral KTi₂(PO₄)₃ is introduced to investigate the related theoretical, structural, and electrochemical properties in K cells. The suggested KTi₂(PO₄)₃ modified by electro-conducting carbon brings about a flat voltage profile at ≈1.6 V, providing a large capacity of 126 mAh (g-phosphate)^{−1}, corresponding to 98.5% of the theoretical capacity, with 89% capacity retention for 500 cycles. Structural analyses using electrochemical performance measurements, first-principles calculations, ex situ X-ray absorption spectroscopy, and *operando* X-ray diffraction provide new insights into the reaction mechanism controlling the (de)intercalation of potassium ions into the host KTi₂(PO₄)₃ structure. It is observed that a biphasic redox process by Ti^{4+/3+} occurs upon discharge, whereas a single-phase reaction followed by a biphasic process occurs upon charge. Along with the structural refinement of the electrochemically reduced K₃Ti₂(PO₄)₃ phase, these new findings provide insight into the reaction mechanism in Na superionic conductor (NASICON)-type KTi₂(PO₄)₃. The present approach can also be extended to the investigation of other NASICON-type materials for potassium-ion batteries.

K₃V₂(PO₄)₃,^[12] KVPO₄F,^[13] KVOPO₄,^[14] and KVP₂O₇,^[15] were demonstrated as 4 V class cathode materials for KIBs. Although they have relatively smaller capacity than layered compounds, their resulting operation voltages were higher than those of layered materials due to the inductive effect associated with the intrinsic nature of P–O and S–O bonds polyanion compounds, which provides good cycling performances over cycles. Prussian blue (PB) and analogues are of interest because of their capacity and cyclability for long term, which are supported by their reversible structural change during de/potassiation.^[16–19] In particular, Prussian white, K_{1.75}Mn[Fe(CN)₆]_{0.93}·0.16H₂O delivered a high discharge capacity of 130 mAh g^{−1} with average operation voltage of 3.75 V.^[19] However, this analogues need to further minimize the amount of intercalated water molecules and defects in the structure and to improve durability of the materials during reaction in electrolyte high voltage. Density of the material is another important issue to increase the energy density of the active materials.

1. Introduction

K-ion batteries (KIBs) are a cost-effective alternative to lithium-ion batteries (LIBs), which are commonly used in portable electronic devices, energy storage systems, and electric vehicles. The increasing number of studies devoted to materials for KIBs confirms the growing interest in this type of battery.^[1–6] The main challenge for KIBs is the limited number of intercalation electrode materials that can accommodate large K⁺ while providing high capacity and stable cyclability.^[7–10] Potassiated transition metal polyanions, such as KMSO₄F (M: Fe, Ni, and Co),^[11]

materials during reaction in electrolyte high voltage. Density of the material is another important issue to increase the energy density of the active materials.

Na superionic conductor (NASICON) ATi₂(PO₄)₃ (A: Li, Na, or K) compounds provides an interesting model for revealing the relationship between the electrochemical properties, cation size, and structural transformations during cycling, although the corresponding operation voltage is lower than the above-mentioned polyanion and PB systems. Structural transformations of the lithium and sodium analogues of ATi₂(PO₄)₃ (A: Li or Na) have been actively studied in lithium- and sodium-ion cells, respectively.^[20–22] However, only a few studies on their potassium analogues (A: K) have been performed, and these works mainly cover the electrochemical performance of the materials without extension of their scopes toward theoretical and mechanistic aspects.^[23–25]

The possibility of using KTi₂(PO₄)₃ as a potassium intercalation electrode in potassium cells has been demonstrated in some previous works. In 2016, Han et al. first showed that potassium ions can be de/intercalated into the KTi₂(PO₄)₃ structure in the voltage range of 1.2–2.8 V, although carbon-coated KTi₂(PO₄)₃ delivered a relatively small capacity of ≈75 mAh g^{−1} with a sloppy voltage plateau at 1.7 V.^[23] Later work performed by Du et al.^[24] introduced a KTi₂(PO₄)₃@C nanocomposite, which was active in the voltage range of 0–3 V with a high

N. Voronina, J. H. Jo, Prof. J. Kim, Prof. S.-T. Myung
Department of Nano Technology and Advanced Materials Engineering
Sejong Battery Institute
Sejong University
Seoul 05006, South Korea
E-mail: jongsoonkim@sejong.ac.kr; smyung@sejong.ac.kr

A. Konarov
National Laboratory Astana
Nazarbayev University
53 Kabanbay Batyr Avenue, Nur-Sultan 010000, Kazakhstan

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202001090>.

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discharge capacity of $\approx 400 \text{ mAh g}^{-1}$. Although short voltage plateaus were evidently observed, the corresponding large capacity was not explained by the $\text{Ti}^{4+}/^{3+}$ and $\text{Ti}^{3+}/^{2+}$ redox pairs, which can theoretically contribute to 256 mAh g^{-1} assuming insertion/extraction of 4 K per formula unit. The overcapacity can likely be attributed to the additional redox-active reaction of carbon that was used as a conductive agent and carbon that was used to form the $\text{KTi}_2(\text{PO}_4)_3$ @C nanocomposite. $\text{KTi}_2(\text{PO}_4)_3$ was also studied in a water-in-salt electrolyte (concentrated potassium acetate).^[25] Although the delivered capacity was 53 mAh g^{-1} ($\approx 0.83 \text{ K}$ per formula unit), the chemistry in the aqueous solution exhibited better kinetics and lower potential hysteresis than that in the aprotic one.

Although there are few works on $\text{KTi}_2(\text{PO}_4)_3$ as a potassium insertion electrode,^[23–25] the related properties of the material is still remained as veiled that motivates us to investigate $\text{KTi}_2(\text{PO}_4)_3$ compound, as follows; namely, i) structural transformation in the $\text{KTi}_2(\text{PO}_4)_3$ during de/potassiation, ii) detailed structural information after potassiation such as K-rich $\text{K}_3\text{Ti}_2(\text{PO}_4)_3$ phase, iii) electrochemical reaction mechanism, and iv) diffusion barrier energy for K^+ ion migration in the $\text{KTi}_2(\text{PO}_4)_3$. For the first time, we herein report those unknown properties of $\text{KTi}_2(\text{PO}_4)_3$ under KIB system. Because of its low electric conductivity, surface modification by an electro-conducting medium is necessary for further investigation of the material, particularly for fully depotassiated states. As anticipated, the present carbon-coated $\text{KTi}_2(\text{PO}_4)_3$ electrode exhibited a flat voltage plateau at $\approx 1.6 \text{ V}$ with a discharge capacity of 126 mAh g^{-1} (12.8 mA g^{-1}), corresponding to 98.5% of the theoretical

capacity. The electrode showed good performance, with 89% capacity retention for 500 cycles. Because the capacity retention is associated with structural stability, the detailed redox mechanism of $\text{KTi}_2(\text{PO}_4)_3$ was elucidated using first-principles calculations, *operando* X-ray diffraction (*o*-XRD), and ex situ X-ray absorption near-edge spectroscopy (XANES) analyses. *O*-XRD analysis confirmed the occurrence of an asymmetrical reaction process on charge and discharge. Specifically, a single-phase reaction was observed for $2.54 < x < 3$ in $\text{K}_x\text{Ti}_2(\text{PO}_4)_3$ followed by a biphasic reaction for $1 < x < 2.54$ in $\text{K}_x\text{Ti}_2(\text{PO}_4)_3$ during charge, whereas a biphasic reaction was dominant during discharge. Ex situ XANES proved that the redox reaction involves a $\text{Ti}^{4+}/^{3+}$ redox pair along with insertion of 2 K per formula unit of $\text{KTi}_2(\text{PO}_4)_3$ in the voltage range of 1–4 V. The structure of the obtained $\text{K}_3\text{Ti}_2(\text{PO}_4)_3$ was refined in the $R\bar{3}c$ space group for the first time, to the best of our knowledge. The above findings suggest that the acceptable level of electrode performance can be attributed to the outstanding structural stability of $\text{KTi}_2(\text{PO}_4)_3$ and is also applicable to other NASICON-type compounds.

2. Results and Discussion

The formation of the pure NASICON-type phase of $\text{KTi}_2(\text{PO}_4)_3$ (KTP) and carbon-coated $\text{KTi}_2(\text{PO}_4)_3$ (KTP/C) was confirmed by the Rietveld refinement of XRD data that was performed by assuming the $R\bar{3}c$ space group (Figure 1a,b). The calculated unit cell and lattice parameters were constant for KTP and KTP/C, and the obtained values were consistent with the

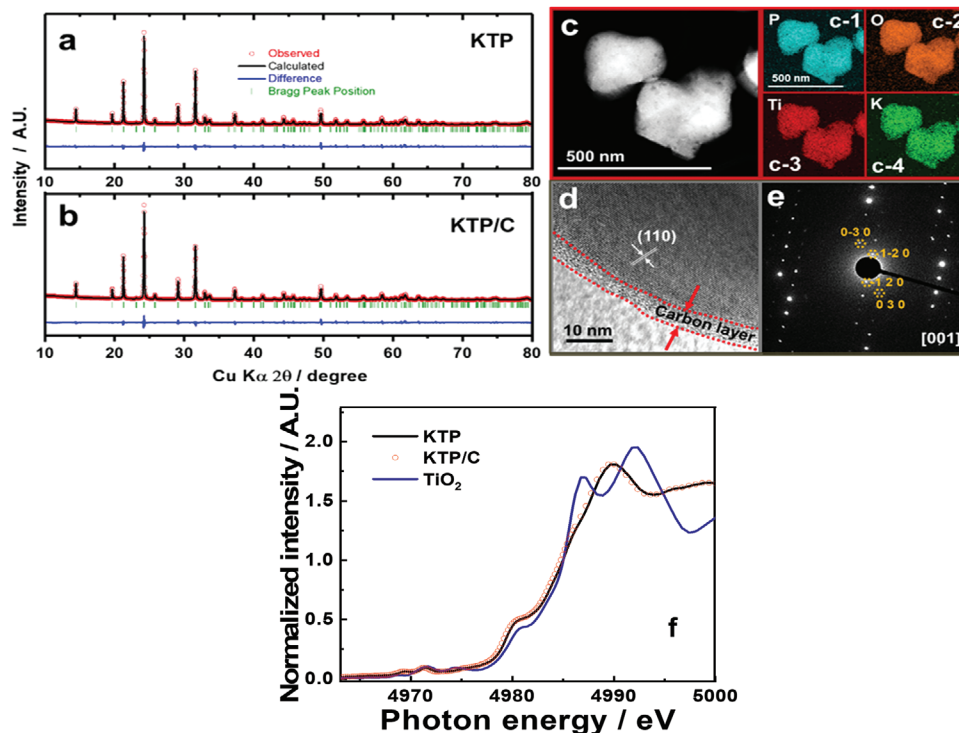


Figure 1. The Rietveld refinement results of XRD data for a) KTP and b) KTP/C; c) TEM image of KTP and corresponding EDS mappings of c-1) P, c-2) O, c-3) Ti, and c-4) K; d) HR-TEM image of KTP/C, e) corresponding [001] SAED pattern, and f) X-ray absorption near-edge structure spectroscopy data of KTP and KTP/C.

results reported in the literature.^[23] This finding indicates that the additional carbon coating treatment did not affect the structure of KTP (Figure 1a,b and Table S1, Supporting Information). The particle sizes and morphology of the as-synthesized KTP and KTP/C were examined using TEM and SEM, as shown in Figure 1c–e and Figure S1 (Supporting Information). The average primary particle sizes of KTP and KTP/C were estimated to be ≈ 200 nm, and the K, Ti, P, and O elements were homogeneously distributed throughout the particles (Figure 1c and Figure S1, Supporting Information). High-resolution TEM analysis revealed that the thickness of the carbon layer was ≈ 5 nm, and the estimated d -spacing was 0.42 nm for the (110) planes, which agrees with the $d_{110} = 0.418$ nm XRD result for rhombohedral KTP (Figure 1d). The corresponding selected-area electron diffraction (SAED) pattern of the KTP structure indicated that the observed structure is identical to the rhombohedral structure of KTP/C that was refined in Figure 1b (Figure 1e). The presence of carbon on the KTP/C surface was further evidenced by Raman spectroscopy (Figure S2, Supporting Information), where the peaks at 1585 and 1345 cm^{-1} can be interpreted as G (sp^2) and D (sp^3) bands derived from the carbon present with KTP powder, respectively. The remaining carbon content in KTP/C (≈ 4.7 wt%) was identified by an elemental analyzer. The presence of carbon on the surface of KTP increased the electrical conductivity from 5×10^{-6} to $7 \times 10^{-3} \text{ S cm}^{-1}$ (Table S1, Supporting Information). The as-synthesized KTP and KTP/C were examined by XANES to understand variation of the oxidation state of Ti before and after the carbon coating. It is clear from the above results, there is no change of the average oxidation state of Ti before and after the carbon coating. We also examined the surface chemical state of Ti using X-ray photoelectron spectroscopy (XPS) (Figure S3, Supporting Information). Similar to the XANES data, the binding energies related to TiO_2 are identical to be 4+ for both KTP and KTP/C. The above results indicate that the oxidation state of Ti for both for KTP and KTP/C did not vary even after the carbonization of pitch at 750 °C for 2 h in Ar atmosphere.

The synthesized KTP and KTP/C were first discharged (reduced) and then charged (oxidized) in galvanostatic mode in the voltage range of 1–4 V at a rate of 0.1C (1C = 128 mA g^{-1}) (Figure S4, Supporting Information). A flat voltage plateau was observed at 1.56 V for both KTP and KTP/C on discharge; however, the plateau was much shorter for the KTP than for the KTP/C, which may be attributed to the lower electrical conductivity of KTP. Even after the test at a current of 1.28 mA g^{-1} (0.01 C), the KTP electrode delivered still smaller than that of the KTP/C electrode tested at 12.8 mA g^{-1} (0.1 C) (Figure S5, Supporting Information). The KTP/C delivered a high discharge capacity of 145 mAh g^{-1} at the first discharge, whereas the discharge capacity of the carbon-free KTP electrode was limited to 75 mAh g^{-1} at 0.1C-rate (Figure S4, Supporting Information). Similar results are typically observed for LiFePO_4 ,^[26] $\text{Li}_4\text{Ti}_5\text{O}_{12}$,^[27] and $\text{Na}_3\text{V}_2(\text{PO}_4)_3$,^[28] which delivered higher capacity after carbon coating. The increased electrical conductivity significantly improves the charge transfer kinetics of coupled K^+/e^- migration during electrochemical process. On charge, the KTP and KTP/C materials presented voltage profiles with discontinuity, resulting in two-step voltage plateaus, as

observed in the dQ/dV and galvanostatic curves in Figure 4a,b. For the second discharge, the plateau at 1.56 V shifted to slightly higher voltage, 1.61 V, and the capacity became stabilized at 30 and 126 mAh g^{-1} and for KTP and KTP/C, respectively. The irreversible capacity observed at the first cycle may be related to reductive decomposition of the electrolyte or the formation of a solid-electrolyte interphase (SEI) layer. Note that the capacity obtained at the second discharge, 126 mAh g^{-1} , corresponds to movement of 2K^+ per formula unit of KTP/C. The theoretical capacity of $\text{KTi}_2(\text{PO}_4)_3$ is 128 mAh g^{-1} ; thus, the present KTP/C composite was able to deliver 98.5% of the theoretical value. A continuous cycling test of the KTP/C electrode revealed a capacity of 112 mAh g^{-1} after 500 cycles at 0.1C, corresponding to 89% capacity retention compared with that of the second cycle (Figure 2a,b). Furthermore, the voltage variation was negligible during long-term operation. The high-rate performance of KTP was significantly improved through carbon coating (Figure 2c,d).

Galvanostatic intermittent titration technique (GITT) measurements were used to deduce the K^+ diffusion coefficients (D_{K^+}) in KTP/C material during charge/discharge (Figure S6, Supporting Information). In consideration of the GITT results with the open-circuit voltage (OCV) points at equilibrium states, the observed voltage plateaus on discharge and charge may imply that the related process would be a typical bi-phasic reaction. The D_{K^+} values varied in the range of 10^{-11} – $10^{-13} \text{ cm}^2 \text{ s}^{-1}$ during discharge and in the range of 10^{-12} – $10^{-14} \text{ cm}^2 \text{ s}^{-1}$ during charge. These results imply that the insertion of K^+ proceeded faster into the KTP structure than K^+ extraction from the structure. The values are close to those determined in the previous work of Xu et al.^[23] and are also comparable to those of other compounds with polyanion structures.^[11–15] In addition, the measured K^+ diffusivity ranged from 10^{-14} to $10^{-17} \text{ cm}^2 \text{ s}^{-1}$ for the carbon-free KTP. This improvement in the electrode performance is thus considered to mainly arise from the increased electrical conductivity of KTP resulting from the electroconducting carbon coating.

Prediction of the possible vacant sites for K^+ ions in $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$ is important because this information indicates the capability of $\text{K}_1\text{Ti}_2(\text{PO}_4)_3$ for the storage of numerous K^+ as a promising electrode material for KIBs. Thus, we performed bond-valence-energy-landscape (BVEL) analyses using the structural information for $\text{K}_1\text{Ti}_2(\text{PO}_4)_3$ verified through the Rietveld refinement. As shown in Figure 3a,b, it was predicted that K^+ ions can be intercalated in the two vacant sites of the $\text{K}_1\text{Ti}_2(\text{PO}_4)_3$ structure, and the Wyckoff positions and (x, y, z) coordinates on the possible K^+ sites are $6b$ (0, 0, 0) and $18e$ (0.333, 0, 0.25) for fully potassiated $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$. Moreover, the confirmed K^+ sites in the $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$ structure are similar to the atomic positions of Na ions in rhombohedral $\text{Na}_3\text{Ti}_2(\text{PO}_4)_3$,^[21] which indicates the high accuracy of the K^+ atomic positions in fully potassiated $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$.

Based on the detailed structural information of fully potassiated $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$, we performed first-principles calculation and predicted various theoretical electrochemical properties of $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$ during K^+ de/intercalation. Numerous K^+ /vacancy configurations for the various $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$ compositions ($0 \leq x \leq 2$) were generated using the CASM code.^[29] After verifying the formation energies of all the configurations of the

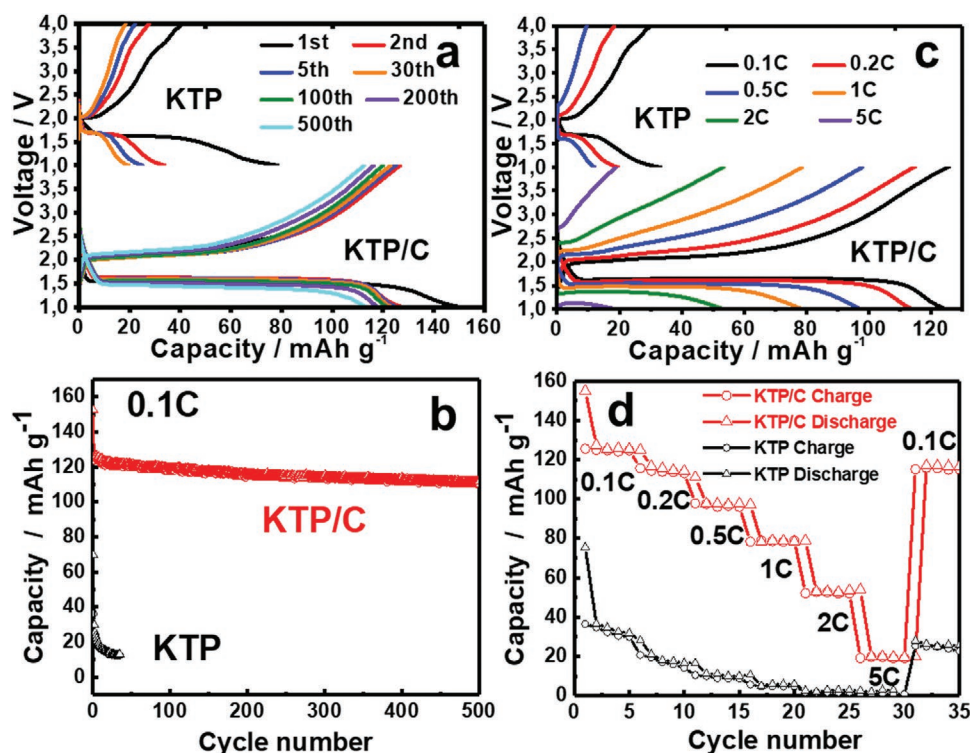


Figure 2. Electrochemical performance of KTP and KTP/C in the voltage range of 1–4 V: a) capacity versus voltage curves for KTP and KTP/C electrodes at 0.1C (12.8 mA g⁻¹) and b) corresponding cycling performance; c) capacity versus voltage curves at currents of 0.1–5C and d) corresponding cycling performance at different currents.

$K_{1+x}Ti_2(PO_4)_3$ compositions, we calculated the theoretical redox potentials for each composition using Equation (1)

$$V = - \frac{E[K_{x_2}Ti_2(PO_4)_3] - E[K_{x_1}Ti_2(PO_4)_3] - (x_2 - x_1)E(K)}{(x_2 - x_1)F} \quad (1)$$

As shown in Figure 3c, it was predicted that 2 mol K⁺ ions are de/intercalated in the $K_{1+x}Ti_2(PO_4)_3$ structure with an average redox potential of ≈ 1.69 V (vs K⁺/K). The predicted electrochemical properties of $K_{1+x}Ti_2(PO_4)_3$ were well matched with the experimentally measured charge/discharge curves of $K_{1+x}Ti_2(PO_4)_3$ (Figure 3d), indicating the high accuracy of our first-principles calculation. In particular, the flat voltage range in the charge/discharge curves indicates the two-phase reaction of $K_{1+x}Ti_2(PO_4)_3$ during charge/discharge under the KIB system, which is closely related to the instability of intermittent phases among $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$ predicted through first-principles calculation (Figure 3c). In addition, we used the nudged elastic band (NEB) method based on first-principles calculations to predict the theoretical K⁺ diffusion motion and the activation barrier energy required for K⁺ diffusion in the $K_{1+x}Ti_2(PO_4)_3$ structure. As shown in Figure 3e,f, we verified that ≈ 620 meV is required for K⁺ diffusion along the K1–K2 pathway, which indicates that K⁺ diffusion in the $K_{1+x}Ti_2(PO_4)_3$ structure is not fast compared to the value for Na⁺ in and $Na_{1+x}Ti_2(PO_4)_3$ (≈ 485 meV).^[22] It is likely that the large ionic size of K⁺ (1.38 Å) relative to Na⁺ (1.02 Å) would be a possible reason for the poor rate capability comparing with the carbon-coated $NaTi_2(PO_4)_3$. This finding implies that special treatments, such

as carbon-coating or nanosizing, are required, to improve the power capability of $K_{1+x}Ti_2(PO_4)_3$ as an electrode for KIBs. This prediction for K⁺ diffusion in the $K_{1+x}Ti_2(PO_4)_3$ structure is well matched with experimental results showing the highly enhanced power capability of $K_{1+x}Ti_2(PO_4)_3$ after carbon coating (Figure 2d).

To understand the mechanistic reaction of KTP/C upon the potassium de-/intercalation processes in detail, *operando* XRD analysis was performed at 0.1C within the potential window of 1–4 V (Figure 4). Upon discharge for the (104), (113), (024), (116), and (300) planes, the resulting XRD peaks at 19.6°, 24.2°, 29.1°, 31.6°, and 37.3° (2 θ), respectively, were gradually weakened; however, immediately, new peaks emerged at 19.3°, 23.0°, 27.7°, 30.9°, and 34.6° (2 θ) because of the formation of the potassium-rich phase $K_{1+x}Ti_2(PO_4)_3$ (Figure 4a and area I in Figure 4b). When the cell was fully discharged to 1 V, only the peaks of the potassium-rich $K_3Ti_2(PO_4)_3$ phase could be observed. The crystal structure of the $K_3Ti_2(PO_4)_3$ phase was able to be refined with $R\bar{3}c$ symmetry with K⁺ ions occupying both 6b and 18e Wyckoff sites and lattice parameters $a = 9.0075(7)$ Å, $c = 22.6572(19)$ Å (Figure 4c). Table 1 summarizes the lattice parameters and refined atomic coordinates of $K_3Ti_2(PO_4)_3$ in the $R\bar{3}c$ space group.

Notably, we observed an abnormal phase transition of $K_{1+x}Ti_2(PO_4)_3$ during the charging process in the voltage range of 1.0–2.1 V (vs K⁺/K), which corresponds to a solid-solution reaction accompanying a linear shift of the XRD peaks during charge. In area II in the voltage profile (Figure 4b), the XRD peaks corresponding to the K-rich $K_3Ti_2(PO_4)_3$ phase

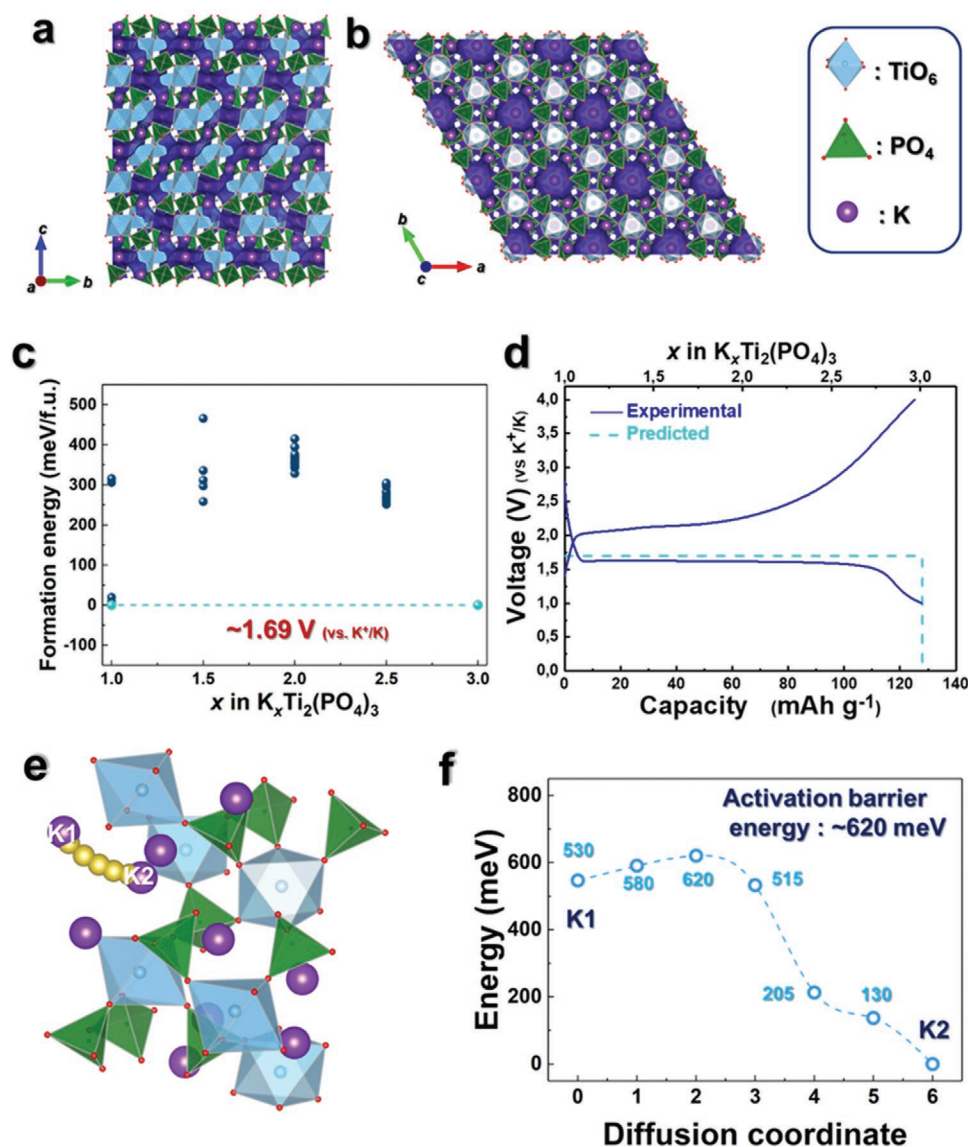


Figure 3. Crystal structures of $K_1Ti_2(PO_4)_3$ with possible vacant sites for K^+ ions along a) bc plane and b) ab plane predicted through BVLE analyses. c) Convex-hull plot for various $K_{1+x}Ti_2(PO_4)_3$ configurations ($0 \leq x \leq 2$) with theoretical redox potential. d) Comparison of predicted redox potential of $K_{1+x}Ti_2(PO_4)_3$ and its experimentally measured charge/discharge curves for the second cycle. e) Predicted K^+ diffusion motion in $K_3Ti_2(PO_4)_3$ structure and f) predicted activation barrier energy for K^+ diffusion in $K_3Ti_2(PO_4)_3$.

progressively shifted to higher angles upon reaching the intermediate phase of $K_{2.54}Ti_2(PO_4)_3$, as derived from electrochemical data. The relative intensity of the $K_{2.54}Ti_2(PO_4)_3$ phase gradually vanished over 2.1 V, and further extraction of potassium resulted in the formation of the potassium-deficient $K_1Ti_2(PO_4)_3$ in area III (Figure 4b). As a result, the related reaction can be summarized as a two-potassium extraction through a single-phase reaction to reach $K_{2.54}Ti_2(PO_4)_3$ from $K_3Ti_2(PO_4)_3$ followed by a biphasic reaction between the intermediate $K_{2.54}Ti_2(PO_4)_3$ and potassium-deficient $K_1Ti_2(PO_4)_3$ phase. Similar phenomena were also observed in $KVPO_4F$ by Antipov et al.^[13] and $NaFePO_4$ as reported by Rojo et al.^[30]. The Rietveld refinement of the potassium-deficient $K_1Ti_2(PO_4)_3$ phase was performed at the end of charge assuming the $R\bar{3}c$ space group (Figure 4d).

To compare the structural change with the results obtained by first principle calculation during K^+ de/intercalation in detail, we visualized the crystal structures of $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$. As shown in Figure 4e, it was verified that all the lattice parameters of $K_{1+x}Ti_2(PO_4)_3$, such as a , b , and c , increase during 2 mol K^+ intercalation, and the predicted total volume change is $\sim 11.68\%$, which is similar to the Rietveld refinement results for pristine $K_1Ti_2(PO_4)_3$ (volume = ~ 1397.13 Å³) (Figure 4c) and fully potassiated $K_3Ti_2(PO_4)_3$ (volume = ~ 1592 Å³) (Figure 4d). The subsequent second discharge resulted in a concomitant biphasic transformation between the K-deficient $K_1Ti_2(PO_4)_3$ and K-rich $K_3Ti_2(PO_4)_3$ phases without the appearance of neither an intermediate phase nor a single-phase reaction (Figure 4a,b).

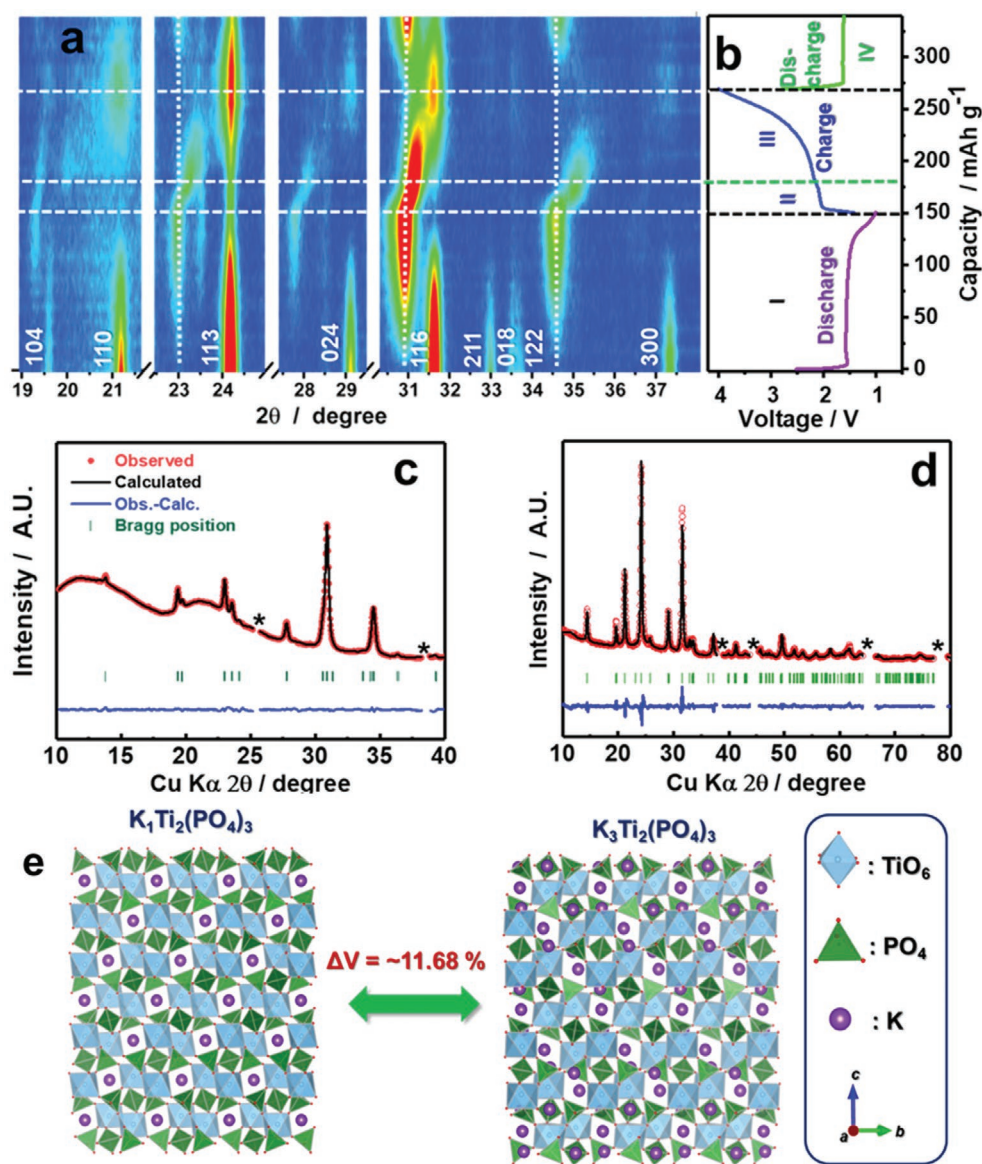


Figure 4. Structural evolution of KTP/C: a) 2D image of *operando* XRD data during first discharge/first charge/beginning of second discharge and b) corresponding voltage profile for *operando* XRD measurement; c) in situ Rietveld refinement results of XRD data for KTP/C electrode after full discharge to 1 V; d) ex situ Rietveld refinement results of XRD data for KTP/C electrode after full charge to 4 V. (*—break from peaks of Al current collector). e) Comparison of predicted crystal structures of $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$.

Table 1. Structural parameters of the Rietveld refinement for $K_3Ti_2(PO_4)_3$ obtained from in situ XRD data.

Atom	Wyckoff position	x	y	z	B_{iso} [Å]	Occupancy
Ti	12c	0	0	0.152(11)	0.290	1
K1	6b	0	0	0	2.290	0.990(8)
K2	18e	0.372(9)	0	0.25	2.290	0.668(3)
P1	18e	0.694(12)	0	0.25	0.360	1
O1	36f	0.238(4)	0.173(3)	0.301(11)	0.971	1
O2	36f	0.446(4)	0.270(2)	0.235(10)	0.947	1

Space group $R\bar{3}c$ a : 9.0075(7) Å, c : 22.6572(19) Å. R_p : 6.28%, R_{wp} : 5.99%.

It was supposed that the abnormal phase reaction of $K_{1+x}Ti_2(PO_4)_3$ during K^+ de/intercalation results from the difference in electronic conductivity between $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$. As shown in Figure 5a,b, we verified the theoretical density of states (DOS) on $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$, which indicated that the band-gap of $K_3Ti_2(PO_4)_3$ is much smaller than that of $K_1Ti_2(PO_4)_3$. Thus, the electronic conductivity of $K_3Ti_2(PO_4)_3$ is remarkably higher than that of $K_1Ti_2(PO_4)_3$. For de/intercalation of ions, such as Li^+ , Na^+ , and K^+ , in the crystal structure, sufficiently fast electron migration is required; thus, the difference of electron migration can affect the electrochemical performance of electrode materials. In particular, it has been reported that ionic diffusion of K^+

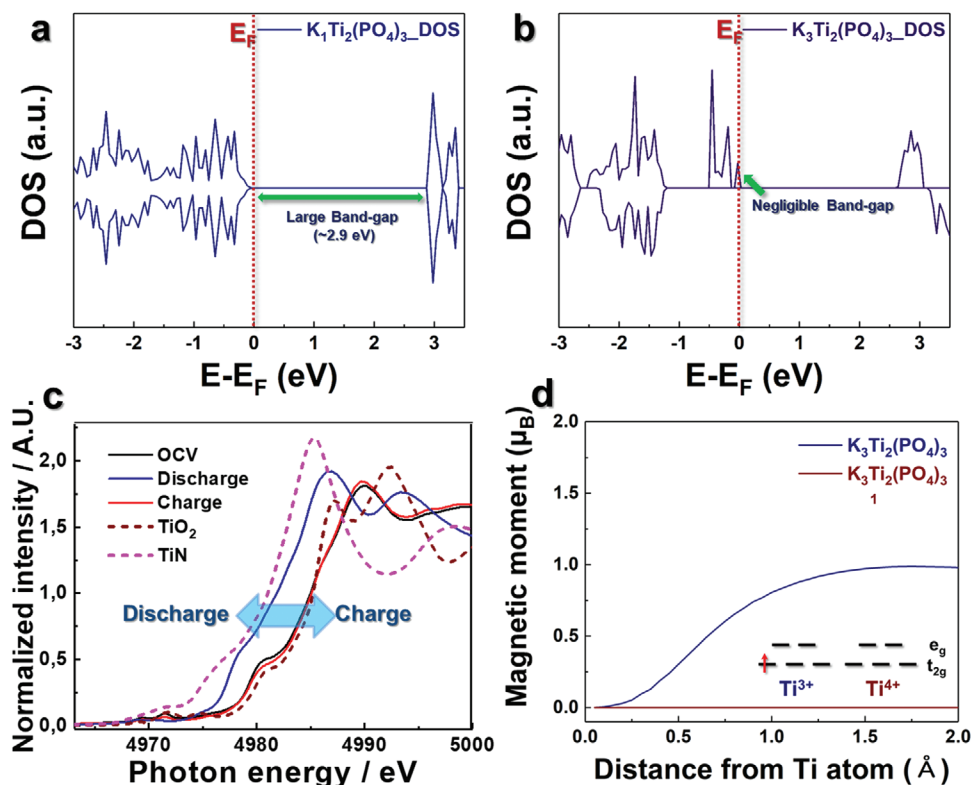
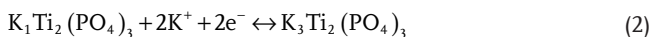


Figure 5. DOS of a) $K_1Ti_2(PO_4)_3$ and b) $K_3Ti_2(PO_4)_3$. c) XANES spectra of Ti K-edge measured at OCV, end of discharge (1 V) and charge (4 V) for KTP/C electrode and references (TiO_2 and TiN). d) Integrated spin moments on Ti ion of $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$.

is more difficult than that of Li^+ or Na^+ because of the larger ionic radius of K^+ , which implies that electron conductivity has a larger effect on the ionic diffusion of K^+ than that of Na^+ or Li^+ . We concluded that the difference in electron conductivity between $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$ results in the abnormal phase reaction of $K_{1+x}Ti_2(PO_4)_3$ mixed with solid-solution and two-phase reaction during only charging process, which is different from the phase reactions of $Li_{1+x}Ti_2(PO_4)_3$ and $Na_{1+x}Ti_2(PO_4)_3$ showing only a two-phase reaction.^[20,21]

To understand the reaction mechanism of the redox reaction of $K_{1+x}Ti_2(PO_4)_3$ during K^+ de/intercalation under a K-cell system, we performed ex situ XANES measurements (Figure 5c). Potassiation (discharge) of the cell to 1 V resulted in a shift of the Ti K-edge spectrum toward the lower-photon-energy direction, and the spectrum approached that of the $Ti^{3+}N$ reference. Depotassiation (charge) to 4 V resulted in a shift of the Ti K-edge spectrum back to higher-photon-energy positions, such that the resulting spectrum became identical to that of the initial state of the KTP/C electrode, which is similar to that of the $Ti^{4+}O_2$ reference. It is thus evident that the reversible electrochemical reaction activated by the Ti^{4+}/Ti^{3+} redox couple and associated with the insertion/extraction of 2 K^+ per formula unit of $K_1Ti_2(PO_4)_3$ in the voltage range of 1–4 V leads to the formation of $K_3Ti_2(PO_4)_3$ through a biphasic reaction on discharge and a single-phase reaction followed by the biphasic reaction on charge. The overall reaction of K^+ insertion/extraction can be described as follows



Moreover, the Ti^{3+}/Ti^{4+} redox reaction of $K_{1+x}Ti_2(PO_4)_3$ during K^+ de/intercalation was confirmed through first-principles calculation. As shown in Figure 5d, the integrated spin moments on Ti ions in $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$ were verified to be close to 0 and 1, respectively, which indicates that theoretically stable oxidation states of Ti ions in $K_1Ti_2(PO_4)_3$ and $K_3Ti_2(PO_4)_3$ are +4 and +3, respectively. Thus, through combined studies using first-principles calculation and XANES analyses, we confirmed the Ti^{3+}/Ti^{4+} redox reaction of $K_{1+x}Ti_2(PO_4)_3$ during charge/discharge under the K-cell system.

To understand the benefits of the synergetic effect of the Ti–P–O framework and carbon coating, the KTP/C cell was disassembled after 500 cycles and the structural stability of KTP/C material was analyzed by ex situ XRD and TEM (Figure S7 and Table S4, Supporting Information). The Rietveld refinement of the XRD data indicated similar structural and lattice parameters as those of the fresh KTP/C material. After 500 cycles, the KTP/C electrodes exhibited high crystallinity, as indicated by the SAED pattern along the [110] zone axis, confirming the maintenance of the original KTP structure after cycling. Overall, the results of these distinct electrochemical and structural studies support our view that the stable polyanion Ti–P–O framework together with carbon coating provides high electrical conductivity to the material and improves the electrode performance.

Here, we summarize the results of the mechanistic properties of the NASICON-type $ATi_2(PO_4)_3$ structure ($A = Li, Na, \text{ or } K$) (Figure 6). From the aspect of crystal chemistry, all alkali-poor $A_1Ti_2(PO_4)_3$ ($A = Li, Na, \text{ or } K$) structures are crystallized into

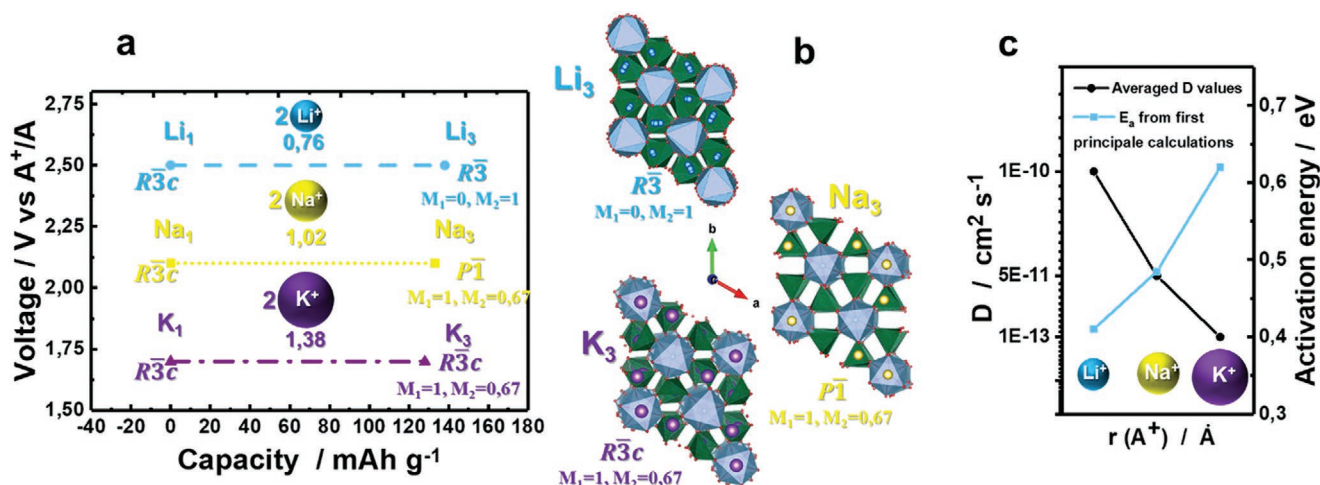


Figure 6. Comparative representation of properties in $\text{ATi}_2(\text{PO}_4)_3 \leftrightarrow \text{A}_3\text{Ti}_2(\text{PO}_4)_3$ system: a) voltage versus theoretical capacity; b) crystal structures of $\text{A}_3\text{Ti}_2(\text{PO}_4)_3$; c) diffusion coefficient values obtained from electrochemical data^[31,32] and activation energies, obtained from first-principles calculations.^[33,34]

a rhombohedral structure with $R\bar{3}c$ space group stabilized in the same three-dimensional framework with spacious channels (Figure 6a). However, the phase-transformation mechanism between the potassium in $\text{KTi}_2(\text{PO}_4)_3 \leftrightarrow \text{K}_3\text{Ti}_2(\text{PO}_4)_3$ is different from that of lithium in $\text{LiTi}_2(\text{PO}_4)_3 \leftrightarrow \text{Li}_3\text{Ti}_2(\text{PO}_4)_3$ ^[20] or sodium in $\text{NaTi}_2(\text{PO}_4)_3 \leftrightarrow \text{Na}_3\text{Ti}_2(\text{PO}_4)_3$.^[21] An asymmetric reaction path is observed on discharge and charge in $\text{KTi}_2(\text{PO}_4)_3$ when two alkali atoms per formula unit are inserted/extracted into/from the structures, unlike the symmetric path in its lithium and sodium analogues. Another difference is that in the $\text{Li}_3\text{Ti}_2(\text{PO}_4)_3$ structure (space group $R\bar{3}$), Li ions are only present within the cavity of the M2 sites (18f). However, in the structures of $\text{Na}_3\text{Ti}_2(\text{PO}_4)_3$ (space group $P\bar{1}$) and $\text{K}_3\text{Ti}_2(\text{PO}_4)_3$ (space group $R\bar{3}c$), Na and K ions prefer to be simultaneously located in both M1 (6b) and M2 (18e) sites (M1 occupancy = 1 and M2 occupancy = 0.667) (Figure 6b). This difference in the distribution of alkali ions within the crystal structure of $\text{A}_3\text{Ti}_2(\text{PO}_4)_3$ (A = Li, Na, K) may be caused by the much smaller size of Li ions compared with that of Na and K ions, and therefore, there is a preference for Li ions to be located only in the smaller M2 cavity. These peculiar characteristics of the NASICON-type $\text{ATi}_2(\text{PO}_4)_3$ structure and the ionic radius differences of Li⁺, Na⁺ and K⁺ may affect the diffusivity and activation energy and, thus, the ionic conductivity of NASICON materials; for example, higher diffusivity and lower activation are observed for $\text{LiTi}_2(\text{PO}_4)_3$ than for $\text{NaTi}_2(\text{PO}_4)_3$ or $\text{KTi}_2(\text{PO}_4)_3$ (Figure 6c).^[32–35] In consideration of the operation voltage of $\text{ATi}_2(\text{PO}_4)_3$ (A: Li, Na, and K), the present $\text{KTi}_2(\text{PO}_4)_3$ can be paired with high voltage cathode materials such as polyanion compounds^[11–15] and PB analogues^[16–19], as 2 V class KIBs. Another possible way to use KTP/C as both anode and cathode in a symmetric KIBs, as we proposed in our earlier work based on $\text{NaTi}_2(\text{PO}_4)_3$ in SIBs,^[34] and Liu et al.^[35] recently demonstrated feasibility of $\text{K}_3\text{V}_2(\text{PO}_4)_3$ in a symmetric KIB cell.

3. Conclusion

In summary, we have demonstrated an effective approach to increase the capacity and improve the high-rate stability of

$\text{KTi}_2(\text{PO}_4)_3$ in aprotic K cells. Carbon coating increases the electrical conductivity of $\text{KTi}_2(\text{PO}_4)_3/\text{C}$ and thus helps achieve a stable cycle life over 500 cycles at 0.1C with capacity retention of 89%. Furthermore, we performed an experimental and theoretical investigation of the structural transformations and ionic transport properties in the NASICON-type $\text{KTi}_2(\text{PO}_4)_3$ structure. Through theoretical analyses based on first-principles calculation, we confirmed that 2 mol K⁺ ions can be reversibly de/intercalated in the $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$ structure with a low redox potential of ≈ 1.69 V (vs K⁺/K) via a two-phase reaction. Moreover, comparison of the DOSs using first-principles calculation revealed that the difference in the intrinsic electronic conductivities of $\text{K}_1\text{Ti}_2(\text{PO}_4)_3$ and $\text{K}_3\text{Ti}_2(\text{PO}_4)_3$ can affect the K⁺ ionic diffusion behavior, which may result in the abnormal solid-solution reaction of $\text{K}_{1+x}\text{Ti}_2(\text{PO}_4)_3$ during charge. Combined *o*-XRD and ex situ XANES analyses indicated that the stable electrochemical reaction in $\text{KTi}_2(\text{PO}_4)_3$ is accompanied by a Ti⁴⁺/Ti³⁺ redox reaction, which is associated with the insertion/extraction of 2 mol K⁺ per formula unit of $\text{KTi}_2(\text{PO}_4)_3$ in the voltage range of 1–4 V (vs K⁺/K). Our findings on the structural transformations and ion migration in $\text{KTi}_2(\text{PO}_4)_3$ will be beneficial for the design of new electrode materials and solid electrolytes with improved performance for KIBs.

4. Experimental Section

Synthesis of KTP and KTP/C: $\text{KTi}_2(\text{PO}_4)_3$ was synthesized through a precipitation process using $\text{Ti}(\text{OC}_4\text{H}_9)_4$, $\text{NH}_4\text{H}_2\text{PO}_4$, and KNO_3 in an ethanol/water solution media. A stoichiometric ratio of the precursors (K:Ti:P = 1:2:3) was dissolved in ethanol/water (2:1 v/v ratio) and stirred for 3 h. Subsequently, the solvent was evaporated at 120 °C, and the resultant was dried in an oven at 120 °C overnight. Finally, the precursor was calcined at 850 °C for 6 h in air. The obtained white product was referred to as KTP. As-synthesized KTP (2 g) was mixed with pitch (8 wt% vs KTP) as a carbon source for surface modification, and the mixture was annealed at 750 °C for 2 h in Ar flow to form KTP/C.

Characterization: XRD patterns were obtained using a PANalytical X'Pert diffractometer with a Cu K α source. The samples were scanned

from 10° to 80° (2θ) with a step size of 0.03° . The FULLPROF program was used for refinement of the crystal structure of the KTP and KTP/C compounds.^[36] The microscopic morphologies were examined using high-resolution TEM (H-800, Hitachi) and scanning electron microscopy (SEM; SU-8010, Hitachi) coupled with energy-dispersive X-ray (EDX) spectrometry. The electrical conductivities of KTP and KTP/C were measured using standard four-point probe characterization (CMT-SR1000, AIT). Raman spectroscopy was employed using an inVia Renishaw Raman spectrometer in the region of $200\text{--}3200\text{ cm}^{-1}$. The amount of carbon in the KTP/C material was estimated using an EA110 elemental analyzer (CE Instruments). Structural studies during cycling were performed using *operando* XRD (X'Pert, PANalytical diffractometer, $18^\circ\text{--}38^\circ$ (2θ)) and ex situ XANES. XANES measurements were performed at nanoprobe beamline 8C of the Pohang Accelerator Laboratory (PAL), South Korea. The obtained spectra were processed using the Athena software package.^[37]

Electrochemical Tests: KTP electrodes were prepared by thoroughly mixing KTP (80 wt%), Super P (10 wt%), and polyvinylidene fluoride (PVDF) binder (10 wt%) in *N*-methyl-2-pyrrolidone (NMP). For the KTP/C electrodes, the ratio was 84:6:10, for which the total carbon content in the composite electrode was balanced to be 10 wt%. The homogeneous mixtures were pasted on aluminum (Al) foil using a doctor blade and were then dried overnight at 110°C under vacuum. The electrochemical performances of the KTP and KTP/C electrodes were measured versus K metal in R2032 coin-type cells. Whatman GF/C glass fiber was used as the separator. 0.5 mol dm^{-3} KPF₆ solution in a mixture of ethylene carbonate (EC) and diethyl carbonate (DEC) (1:1 volume%) was used as the electrolyte. The cells were assembled in an Ar-filled glove box with oxygen and moisture contents below 0.1 ppm. Galvanostatic discharge and charge measurements were conducted in the voltage range of 1–4 V at 25°C . GITT measurements were performed with 30 min discharge or charge and relaxation periods of 1 h at a current density of 0.1C ($\text{IC} = 128\text{ mA g}^{-1}$).

Computational Details: First-principles calculation based on density functional theory (DFT) calculations was performed using the Vienna Ab initio Simulation Package (VASP).^[38] Projector-augmented wave (PAW) pseudopotentials^[39] were applied with a plane-wave basis set for the DFT calculation. The generalized gradient approximation (GGA) and the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional were used.^[40] The GGA+U method^[41] was used for consideration of the localization of the d-orbital in the Ti ions with a U value of the Ti ions of 4.2 eV, which was taken from the literature.^[32,42] All the calculations were performed with an energy cutoff of 500 eV until the remaining force in the system converged to less than $0.03\text{ eV/\AA}$ per unit cell. In addition, nudged elastic band (NEB) calculations^[43] were performed for prediction of the activation barrier for K⁺ diffusion in K₃Ti₂(PO₄)₃. Five intermediate states were considered between the first and final states of a single K⁺ ion diffusion. For the NEB calculation, all the structures were allowed to relax within the fixed lattice parameters.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Keywords

first-principles calculations, insertion, KTi₂(PO₄)₃, potassium-ion batteries

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